

SIMPHIWE MNGADI

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<https://github.com/Simphiwemngadi>

Professional Summary

Detail-oriented mathematical science graduate, prepared for analytical roles. Proven ability to extract and interpret data to support business decisions. Passionate about applying data science and statistical techniques to drive evidence-based decision-making.

Skills

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| • Python | • Microsoft Power BI |
| • SQL | • Microsoft Excel |
| • RStudio | • SAS |

Work History

DATA SCIENCE INTERN, Siyaphumelela, Western Cape

July 2025 to December 2025

- Achieved 86.7% model accuracy by developing an XGBoost classification model to predict student throughput, enabling data-driven academic performance forecasting.
- Conducted exploratory data analysis on large student success datasets to uncover key insights, supporting evidence-based strategies for improving retention and graduation rates.
- Visualised top 10 predictors of graduate throughput using clear analytical charts to inform academic improvement initiatives and guide policy decisions with actionable evidence.

DATA ANALYST INTERN, Cape Peninsula University of Technology, Western Cape

July 2024 to November 2024

- Extracted and transformed data from 5 relational tables using SQL and Python within an ETL pipeline, analysing over 5000 records to identify trend identification through EDA and enhancing decision-making processes with impactful Power BI reports.
- Ensured report accuracy by conducting data cleaning and validation tasks to improve the reliability of insights and support consistent, evidence-based business decisions.
- Collaborated with cross-functional teams to translate analytical findings into actionable recommendations, drive strategic improvements, and optimize organizational performance.

DATA CAPTURE & ENUMERATOR, Golden Arrow Bus Services, Western Cape

August 2022 to November 2022

- Enhanced transport service efficiency by collecting and validating commuter survey data, ensuring accurate insights to optimize scheduling and resource allocation.
- Ensured consistency of datasets throughout the project by implementing rigorous data validation and standardization processes, improving reliability of analysis and supporting evidence-based transport planning.

Education

Postgraduate Diploma in Mathematical Sciences

In Progress

Cape Peninsula University of Technology, Bellville, Western Cape

Advanced Diploma in Mathematical Sciences

December 2025

Cape Peninsula University of Technology, Bellville, Western Cape

Diploma in Mathematical Sciences

December 2024

Cape Peninsula University of Technology, Bellville, Western Cape

Projects

Sales Interactive Dashboard, Personal Project | SQL, Power BI

- Identified products with gross margin below 30% in Q3 2025 using SQL-based analysis to flag profitability risks.
- Built an interactive Power BI dashboard to analyse year-over-year gross margin performance and revenue by product, category, and region.
- Delivered data-driven pricing and discontinuation recommendations to improve overall profitability

Customer Segmentation, Personal Project | Python

- Analysed demographic and economic customer data (age, income, spending score) to understand behavioural patterns across mall customers.
- Conducted univariate and bivariate exploratory data analysis to identify relationships, revealing a positive correlation between income and spending score
- Applied K-Means clustering with elbow-method optimisation to segment customers into 5 distinct customer groups based on income and spending behaviour
- Visualised customer segments and cluster centroids to communicate insights that support targeted marketing and customer strategy decisions

Certifications

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- IBM Data Analysis with Python
 - IBM Databases and SQL for Data Science with Python

References

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- Dr. Farrar T, Acting HOD, +27848596809, CPUT
 - Meyer, J, Mentor, 0219596556, CPUT